BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: 102401

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball in separate OMR Sheet. 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Qi. The number of neutrons in $^{226}_{88}Ra$ is _____ .

(A) 88
(B) 138
(C) 226
(D) 314

Qii. Complete the equation: $^{175}_{78}Pt \rightarrow {}^{171}_{76}Os +$ _____ .

(A) α
(B) β^+
(C) β^-
(D) γ

Qiii. A device that permanently stores and retrieves data on a computer is called:

(A) Flash driver
(B) DVD
(C) Floppy
(D) Hard Disk

Qiv. The output of OR gate is written as X=

(A) A.B
(B) A+B
(C) A.B (whole overlined)
(D) A+B (whole overlined)

Qv. If the magnitude of magnetic field linking to the coil is decreased the magnitude of induced emf:

(A) Increases
(B) Decreases
(C) Remains same
(D) Become zero

Qvi. The resistance of a conductor can be decreased by decreasing its:

(A) Radius
(B) Diameter
(C) Area
(D) Length

Qvii. When two capacitors of 20μF and 30μF are connected in parallel the total capacitance is:

(A) 20μF
(B) 30μF
(C) 50μF
(D) 60μF

Qviii. The refractive index of kerosene oil is 1.44, the speed of light in kerosene oil is:

(A) 3.08×10^{8} m/s
(B) 2.08×10^{8} m/s
(C) 2.26×10^{8} m/s
(D) 3×10^{8} m/s

Qix. If the refractive index of two mediums is same then light ray entering from one medium to another move:

(A) Un deviated
(B) Towards normal
(C) Away from normal
(D) Along the normal

Qx. If the speed of sound in water is 1600 m/s, in air it would be:

(A) 160 m/s
(B) 320 m/s
(C) 640 m/s
(D) 960 m/s

Qxi. A spring vibrates with frequency 2.6Hz which produces a wave having speed 0.962m/s, its wave length is:

(A) 0.25m
(B) 2.5m
(C) 0.37m
(D) 3.7m

Qxii. The part of wave above the equilibrium position is called:

(A) compression
(B) crest
(C) Rarefaction
(D) Trough

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for section B and C is 2 hours and 40 minutes.

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

i By giving one example of each distinguish between pitch and quality.

ii Define audible frequency range and distinguish between infrasonic and ultrasonic.

iii With the help of diagram prove that angle of incidence is equal to angle of reflection.

iv Explain the concept of electric potential.

v Differentiate between Ohmic and non-Ohmic devices and give one example of each.

vi A heater of resistance 300 Ω is connected to main supply for 10 minutes. If 15A current flows through the filament of the heater. What is the heat produced in it?

vii Define A.C generator and name three main components of A.C generator.

viii Define CRO and describe the function of screen in CRO.

ix Describe desktop computer and write its three disadvantages.

x Write nature and ionizing power of β particles.

xi Describe fusion with the help of proton-proton cycle.

SECTION "C"

Maximum Marks: 21

Note: Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) Explain the concept of equilibrium position, inertia and restoring force in case of ball in a bowl executing SHM.

(b) Determine the magnitude of the restoring force on a pendulum bob of mass 140g, which has been pulled to an angle of 10° from the vertical. ($g=9.8 m/s^2$)

QIV.

(a) Describe the role of contact lenses in the correctness of myopia and hyperopia.

(b) A biologist with a near-point distance N= 25cm examines an insect through a lens Whose focal length is 4.6cm. Find the angular magnification when image produced at a) near point b) infinity

QV.

(a) Explain the law proposed by Coulomb about electric force between stationary point charges.

(b) Two point charge q1=6.8μC and q2= 3.5μC are placed at a distance of 10cm. Calculate the Coulomb's force between them.

QVI.

(a) What factors contribute to force acting on current carrying conductor in magnetic field? Explain.

(b) A 0.5m long wire carries a current of 5A, at right angle to uniform magnetic field of 0.04T. Determine the force exerted on the wire.